

16 Apr '26

OS

## Memory Management

Non - Continuous Memory - Paging

Shaurya

|       |                 |                                                                                 |
|-------|-----------------|---------------------------------------------------------------------------------|
| 1 bit | 1 Byte = 8 bits | Word                                                                            |
| 1 bit |                 | $\rightarrow$ 8 bits<br>$\rightarrow$ 16 bits<br>$\rightarrow$ 32 bits (common) |

1 KB  $\rightarrow 2^{10}$  bytes

1 MB  $\rightarrow 2^{20}$  bytes

1 GB  $\rightarrow 2^{30}$  bytes

1 TB  $\rightarrow 2^{40}$  bytes

1 PB  $\rightarrow 2^{50}$  bytes

~~1 X~~

Q 10 bit address  $\rightarrow$  size of location 1 bytes

$\rightarrow 2^{10} \times 1 \text{ bytes}$   
1 KB

Q 44 bit address  $\rightarrow$  size of location 2 bytes

$\rightarrow 2^{44} \times 2 \text{ bytes}$   
 $2^{44} \times 2 \text{ bytes}$   
32 TB

Q 64 GB memory

$\rightarrow$  32 bit address

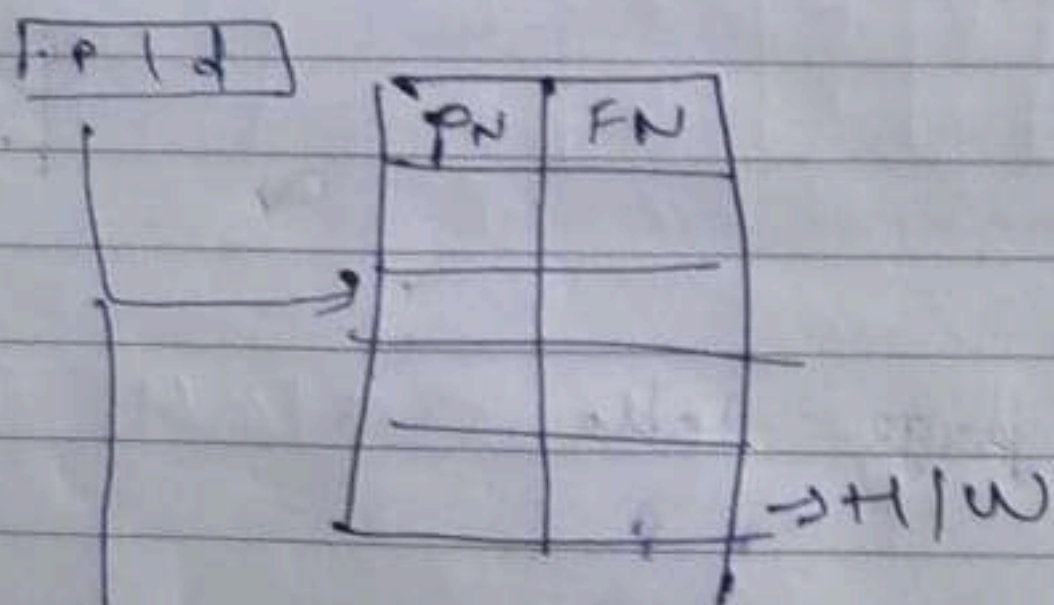
~~$2^{32} \times 16 \text{ GB} \times 2^{30} = 16 \text{ bytes}$~~   
 ~~$4 \times 2^{32}$~~



## Locality of Reference

- ↳ Spatial Locality - space
- ↳ Temporal Locality - time

Translation Lookaside Buffer  $\rightarrow$  stores frame, page number so can access directly.



$\rightarrow$  Page Table (if not found in TLB)

$$EMAT = \overset{\text{Hit Ratio}}{H} (TLB + MMAT) + (1 - H) (TLB + 2MMAT)$$

(with TLB)

$$EMAT = 2MMAT$$

$$MMAT = 400 \text{ microseconds}$$

$$TLBR = 50 \text{ microseconds}$$

$$\text{Hit ratio} = 90\%$$

$$EMAT = \frac{0.9}{100} (50 + 400) + 0.1 (50 + 800)$$

$$= 0.9 \times 450 + 0.1 \times 850$$

$$= 405 + 85$$

$$= 490$$

$$EMAT = 800$$



Ans

$$\frac{6}{10} \times \frac{5}{9} + \frac{4}{10} \times \frac{3}{9}$$
$$\Rightarrow \frac{30+12}{90} \Rightarrow \frac{42}{90} = \frac{1}{1.5}$$

Ans 2

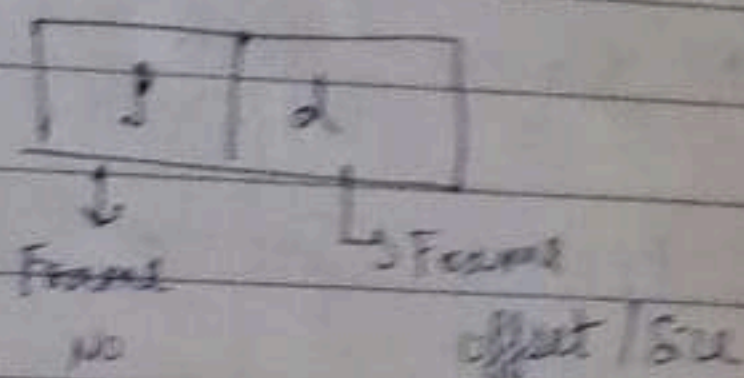
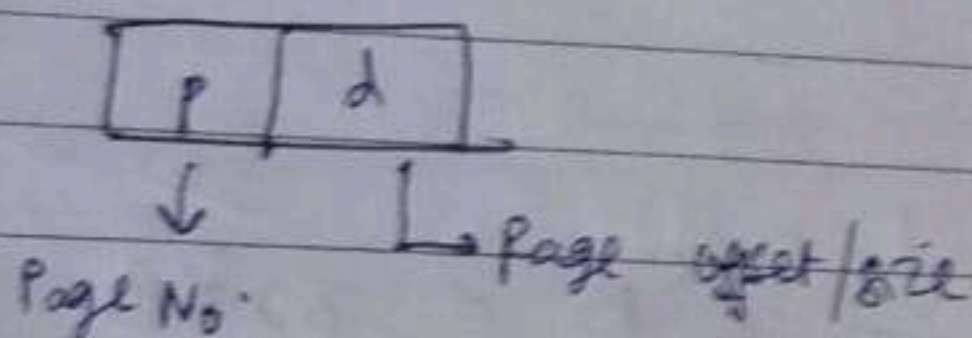
23<sup>rd</sup> Apr '26

OS

Paging / Non continuous memory allocation

Main Memory size  $\rightarrow$  Physical Address space Frames

Logical Address space  $\rightarrow$  Secondary Memory size Pages



Q

$$\text{No. of pages} = 2^{20}$$
$$\text{Page size} = 2B$$

$$\text{Size} \rightarrow 2^{20} \times 2B$$
$$\text{Memory} \quad 2MB$$

Q

$$\text{SM size} = 64GB$$
$$\text{Word addressable memory} = 4B$$

$$\text{No. of pages} = \frac{64GB}{4B} \rightarrow \frac{64 \times 2^{30} B}{4B}$$
$$= 16 \times 2^{30} = 2^{34}$$



Q

L.A = 32 bit

P.A = 16 bit

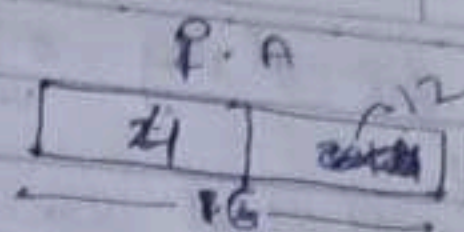
No. of frames = 4 bit

No. of pages = ~~30~~ bit

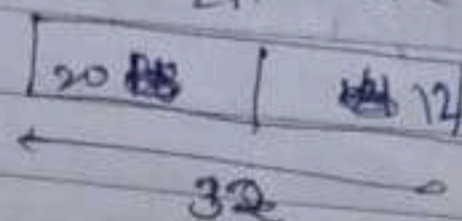
~~18~~ bit

20 bit

|          |       |
|----------|-------|
| Page No. | YOUVA |
| Date     |       |



LA Frame size = ~~16~~ 2<sup>12</sup>



22<sup>nd</sup> Apr '26

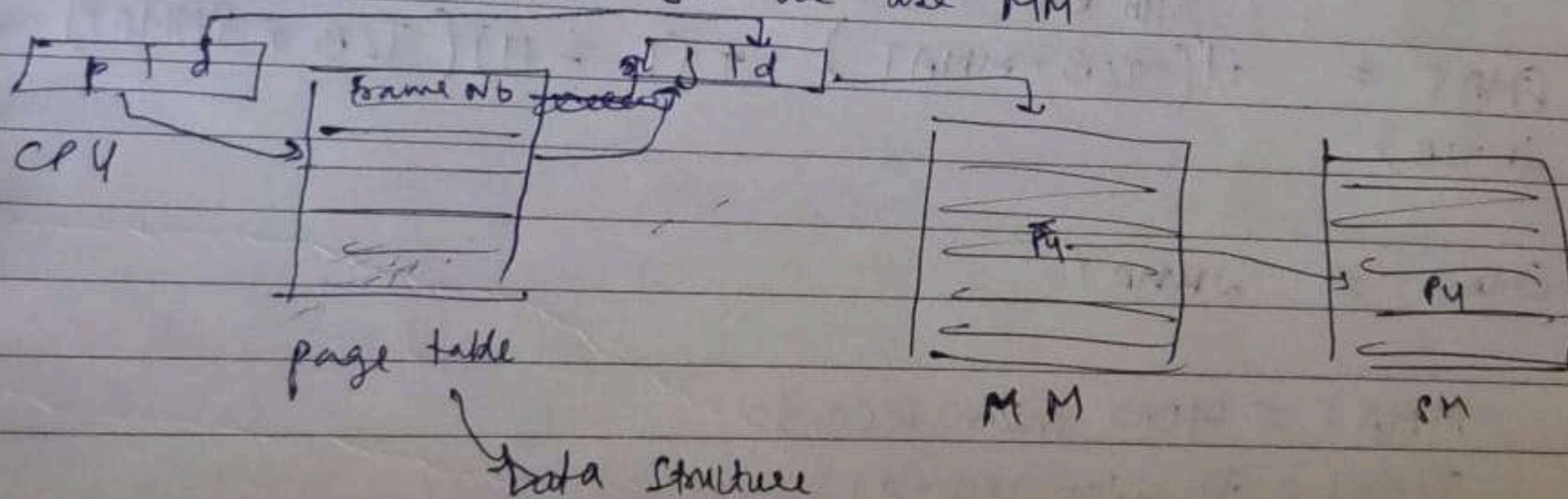
OS

Address Mapping

Every process has its own page table

Address Translation

To access SM we use MM



Effective Memory Access Time = 2 × Main Memory Access Time

Size of Page Table = Number of processes  
= Page Table Entry Size



Main memory size = No. of frames  $\times$  frame size

$$[ \text{Page size} = \text{Frame size} ]$$

Q MM = 32 MB  
 No. of frame = 2048

$$\text{Frame size} = \frac{32 \times 2^{20} \text{ B}}{2048 = 2^{11}} = 2^{14} \text{ B} = 16 \text{ KB}$$

Q MM = 256 MB  
 SM = 512 MB  
 No. of frames = 4096

$$\text{Frame size} = \frac{256 \times 2^{20}}{2^{12}} = 2^{16} \text{ B} = 32 \text{ KB}$$

$$\text{Page size} = 32 \text{ KB}$$

$$\text{No. of Pages} = \frac{2^9 \times 2^{30}}{2^{16}} = 2^{23} = 8192$$

Q Logical  
 Physical A = 64 Bit

$$\text{No. of Pages} = 64$$

$$\text{Page size} = 2^{58}$$

